SCATS LEARNING LAB · STUDY GUIDE

The traffic light that grades its own green time

How SCATS watches one loop per lane, measures how much of each green was really used, and adjusts the next cycle.

Level 1 · Foundation



INSIDE

- Why SCATS counts empty road, not cars
- The degree of saturation, worked by hand in five steps
- Two brains: one deciding every second, one every cycle
- Fixed time, actuated and SCATS raced in SUMO

Beginner-friendly companion to smrahmanbd.github.io/scats-learning-lab

The map: SCATS Foundation

This guide is for complete beginners. You need no traffic engineering and only school maths. By the end you will understand how SCATS, one of the most widely used adaptive traffic systems in the world, measures traffic with a loop at the stop line and turns one number, the degree of saturation, into new signal timings. There are six ideas. Each builds on the one before.

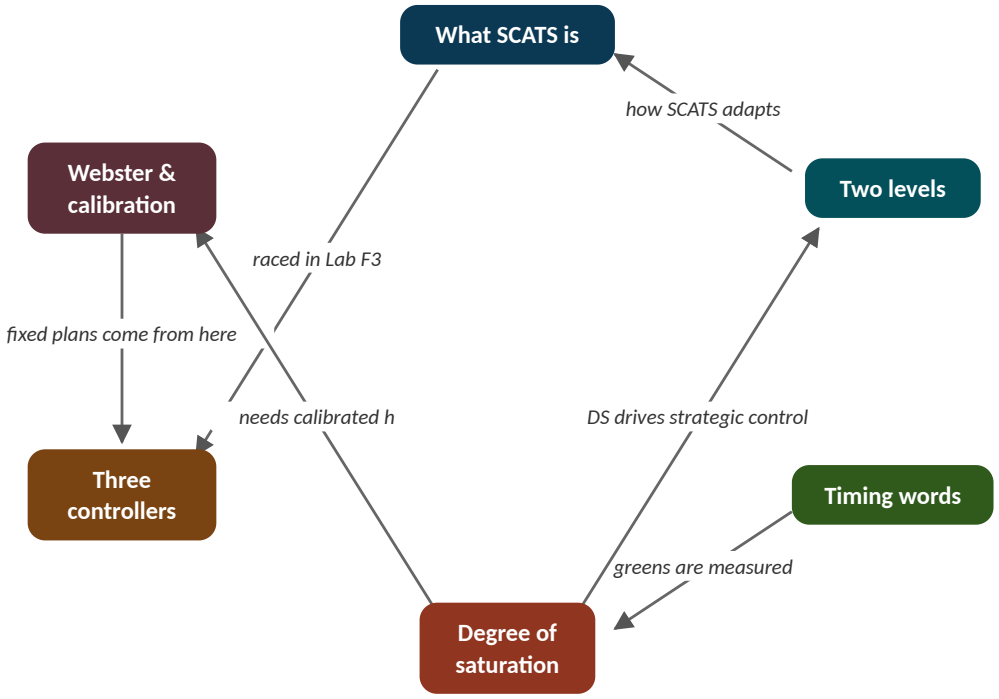


Figure 0. The whole guide on one page. Each box is one concept section; the arrows say how one idea feeds the next. Come back to this page whenever you feel lost.

1. What SCATS is
2. Two levels of control
3. The timing vocabulary
4. Measuring demand: the degree of saturation
5. Fixed time, actuated and SCATS
6. Where fixed-time plans come from

HOW TO USE THIS GUIDE

Read one concept per sitting (20 to 30 minutes). Do every **Simple calculation** with a pencil before reading the answer. Say the **Picture this** scene out loud: the sillier it feels, the better it sticks. The **Recap so far** box at the end of each section grows as you go; reading it is your free revision. When you finish, take the self-quiz, then follow the revision plan on the last page.

What you will be able to do

- Explain what 'adaptive, coordinated and plan-based' mean for SCATS.
- Say which decisions are tactical (every second) and which are strategic (every cycle).
- Use cycle, phase, split, intergreen and offset correctly, and work out the greens of a cycle.
- Compute the degree of saturation (DS) and VK from a loop record by hand.
- Explain why SCATS beat fixed time and roughly tied with actuation at one junction in the SUMO lab.
- Design a fixed plan with Webster's formula and say why SCATS re-solves it every cycle.

Before we start

HOOK

SCATS measures empty road. Instead of counting cars, SCATS adds up the moments when nothing is over the loop during green. Wasted green is easy to see; used green is whatever is left.

HOOK

It runs on a huge scale. SCATS was developed in Sydney from the 1970s and is owned today by Transport for NSW. It is reported to run more than 60,000 intersections in over 200 cities.

HOOK

A truck is handled fairly without being recognised. A truck covers the loop for longer, so it leaves less empty time, exactly as it uses more green. SCATS never needs to know it was a truck.

Words you will meet

You do not need to memorise these now. Each one is explained again when it first matters.

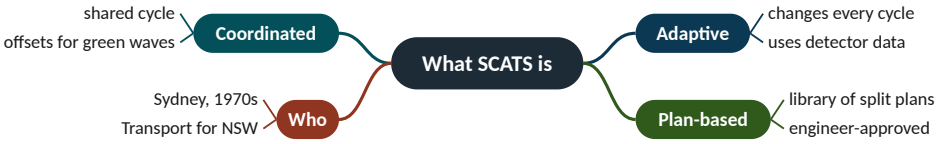
Word	In plain words
Intersection	A place where roads meet and traffic lights take turns.
Phase	A set of movements that get green together, for example 'main road both ways' (phase A).
Cycle	One full turn through all phases, including the changeover times, for example 90 s.
Split	How the green time is shared between phases, for example 60% to A and 40% to B.
Intergreen	Amber plus all-red between two phases. Nobody moves much, so it is 'lost' time.
Offset	When one intersection's cycle starts compared with its neighbour, so a green wave can form.
Stop-line loop	A wire loop in the road just before the stop line that knows when a vehicle is over it.
Degree of saturation (DS)	How much of a green was really used: 1 means fully used, 0.4 means mostly empty road.
Subsystem	A small group of intersections that share one cycle length.

CONCEPT 1 OF 6 · SCATS FOUNDATION

What SCATS is

HOOK

A timing plan written months ago knows nothing about today's crash, concert or rainstorm. SCATS grades every green it just gave, and corrects the next cycle. How can such small corrections keep a whole city moving?



IN PLAIN WORDS

SCATS is like a teacher marking homework every day instead of once a year. Each day she sees what went wrong and adjusts tomorrow's lesson a little. Small daily fixes keep the class on track. *Where the picture breaks:* a teacher can invent any lesson; SCATS mostly chooses and blends from timing plans that engineers approved in advance.

SCATS stands for the **Sydney Coordinated Adaptive Traffic System**. It was developed by the New South Wales Department of Main Roads from the 1970s and is owned and run today by **Transport for NSW**.

Three words describe it. **Adaptive**: timings change automatically from detector data, cycle by cycle. **Coordinated**: neighbouring signals share a cycle and are offset so platoons (bunches of cars) meet green lights. **Plan-based**: SCATS does not invent arbitrary timings; it chooses and blends from a library of split and offset plans approved by engineers.

The core idea is simple. At the end of each cycle SCATS asks: *how well was each green used?* Then it nudges the cycle length and the split for the next cycle. Many small corrections keep the signals in step with real traffic.

SIMPLE CALCULATION · How often does SCATS think?

One intersection running a 90 s cycle for one hour.

1. Cycles per hour = 3600 ÷ 90 = **40**.
2. Strategic decisions (cycle length, split, offset) happen once per cycle: **40** per hour.
3. Tactical checks (should this green end now?) happen every second: 3600 per hour.
4. So for every strategic decision there are 3600 ÷ 40 = **90** tactical checks.

Forty careful decisions an hour, each backed by ninety quick checks. Each correction can be small because another one is coming within two minutes.

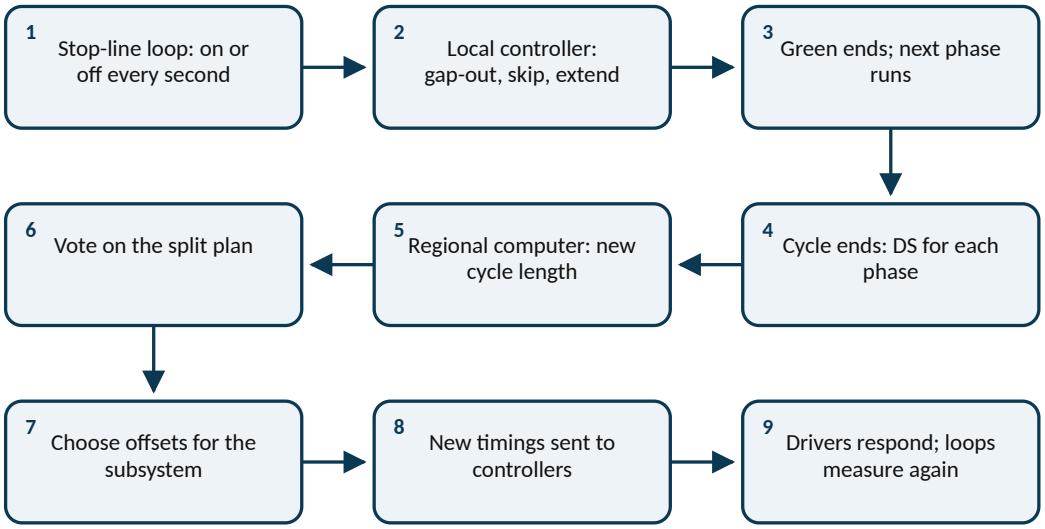


Figure 1. From the loop to the next cycle. What to notice: the loop reports every second, the local controller reacts at once, and the regional computer grades the whole cycle before choosing the next timings. Read left to right, then down.

TRAP

Adaptive does not mean 'anything goes'. SCATS keeps the cycle between a minimum and a maximum, changes it by a few seconds at a time, and picks splits from a small library of approved plans.

PICTURE THIS · STOP 1 OF THE ROUTE

At the start of the bridge a conductor in a huge coat waves a baton at the traffic lights. Instead of one score she flips between four thick music books, and after every bar she scribbles a red mark on the page she just played.

What it encodes: Conductor = coordinated; red marks after every bar = adaptive grading of each cycle; four music books = the library of split plans (plan-based).

TRY IT NOW (about 90 seconds)

An intersection runs a 120 s cycle. How many strategic decisions does SCATS make there in one hour? *Answer in the key at the back.*

RECAP SO FAR

1. **What SCATS is: adaptive (every cycle), coordinated (shared cycle, offsets) and plan-based (approved plans); it grades each green and nudges the next cycle.**

CONCEPT 2 OF 6 · SCATS FOUNDATION

Two levels of control

HOOK

Some decisions must be made in a split second (the road just emptied). Others need the whole picture (traffic is building across the area). SCATS gives each kind of decision its own brain.



IN PLAIN WORDS

A restaurant has waiters and a manager. Waiters react every second: this table is empty, serve the next one. The manager looks at the whole evening and decides how many tables and staff tonight needs. *Where the picture breaks:* a manager can change anything at once; SCATS limits how far the cycle may move in one step so the system never lurches.

Tactical control runs in the roadside controller, every second. Its main tool is **gap-out**: if no vehicle has crossed the loop for a few seconds (the gap setting, 3 s in this lab), the green ends early. It also **skips** a phase nobody is waiting for, and responds to buses and emergency vehicles.

Strategic control runs in a regional computer, once per cycle. It measures the degree of saturation (DS) of each phase, then sets the **cycle length**, the **split plan** and the **offsets** for a group of intersections called a **subsystem**. The cycle is steered so the busiest phase sits near a target DS of 0.9, moving only a few seconds per cycle.

If communications fail, each intersection has a fallback. Full adaptive control is **Masterlink**. **Flexilink** runs time-of-day plans with clocks kept in step. **Isolated** runs local vehicle actuation.

$$\text{desired cycle} = \text{current cycle} \times \text{DS} \div 0.9$$

then limited to a step of +6 s (+9 s if DS ≥ 1) or −6 s, between 40 and 140 s (this lab's values)

SIMPLE CALCULATION · Two cycle-length decisions

The cycle is 80 s. Case 1: the busiest phase had DS 0.99. Case 2: it had DS 0.60.

1. Case 1 desired: 80 × 0.99 ÷ 0.9 = **88 s**, which is 8 s longer.
2. DS is below 1, so the step limit is +6 s: new cycle = 80 + 6 = **86 s**.
3. Case 2 desired: 80 × 0.60 ÷ 0.9 = **53.3 s**, which is 26.7 s shorter.
4. The step limit is −6 s: new cycle = 80 − 6 = **74 s**.

SCATS moves towards the cycle that would give DS 0.9, but only a few seconds per cycle, so one odd cycle cannot swing the timings.

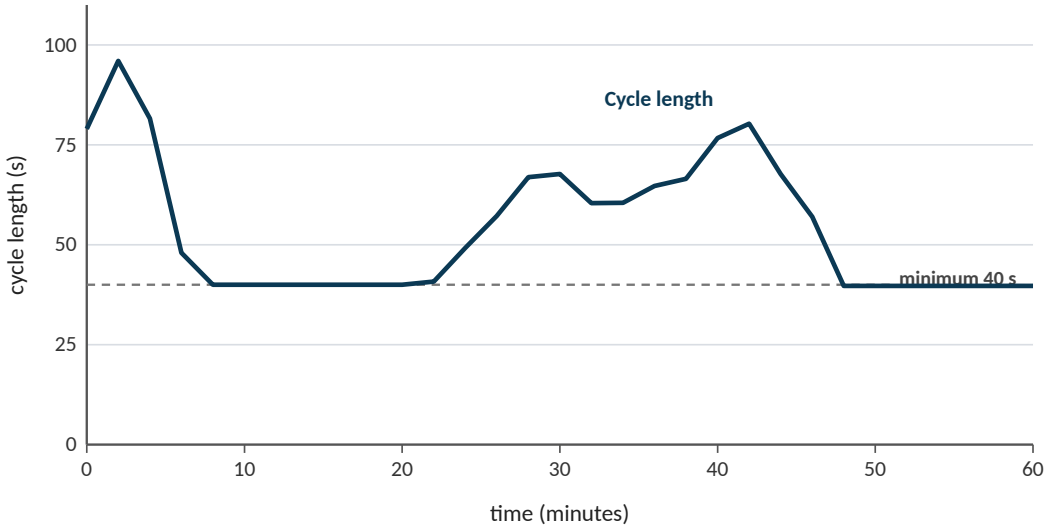


Figure 2. SCATS cycle length over one hour in Lab F3 (SUMO). What to notice: in light traffic the cycle falls to the 40 s minimum; as the peak builds and DS nears 1 it climbs to about 70-80 s; after the peak it falls back. Sampled every 2 minutes from the repository run.

TRAP

Gap-out is tactical, not strategic. Ending a green early because the road emptied happens in the local controller within seconds; it is not a decision of the regional computer.

PICTURE THIS · STOP 2 OF THE ROUTE

Halfway across the bridge a double-decker tram rattles by. Downstairs a driver slaps a big button every single second, shouting 'empty, next!'. Upstairs a captain with a clipboard looks through a telescope and, once a lap, turns a dial by exactly six clicks.

What it encodes: Downstairs driver = tactical control every second (gap-out, skip); upstairs captain = strategic control once per cycle; six clicks = the ±6 s cycle step.

TRY IT NOW (about 90 seconds)

The cycle is 60 s and the busiest phase had DS 1.05. What new cycle does this lab's rule give? *Answer in the key at the back.*

RECAP SO FAR

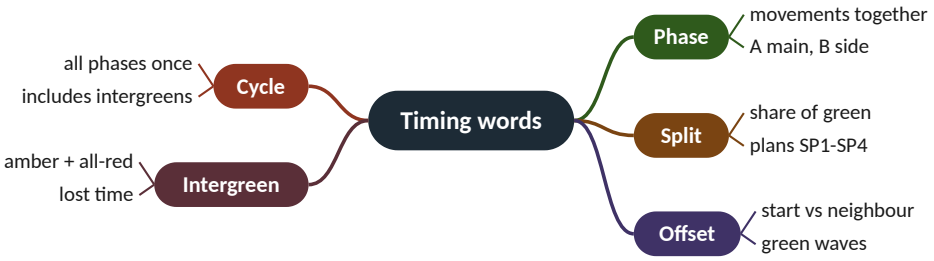
1. **What SCATS is:** adaptive (every cycle), coordinated (shared cycle, offsets) and plan-based (approved plans); it grades each green and nudges the next cycle.
2. **Two levels: tactical control every second in the local controller (gap-out, skip); strategic control every cycle in the regional computer (cycle towards DS 0.9, splits, offsets).**

CONCEPT 3 OF 6 · SCATS FOUNDATION

The timing vocabulary

HOOK

Every signal plan in the world is described with the same five words. Learn them once and every traffic engineering conversation becomes readable.



IN PLAIN WORDS

A cycle is a pizza. Each phase gets a slice; the split says how big each slice is. The crust between slices is the intergreen: it comes with every slice and nobody really eats it. *Where the picture breaks:* a bigger pizza has the same amount of crust per slice, which is exactly why longer cycles waste a smaller share of time.

A **phase** is a set of movements that may run together; here A is the main road and B the side road. The **cycle** (SCATS calls its length RL) is the time for every phase to run once, including the changeovers.

Between two phases comes the **intergreen**: amber plus all-red. Traffic barely moves, so it is **lost time**. Whatever is left of the cycle is green, shared between phases by the **split**. SCATS keeps a small library of split plans (SP1 to SP4 in this lab).

The **offset** is when a cycle starts at one intersection compared with a reference one. Good offsets make green waves; the Advanced level covers them.

$$\text{total green} = \text{cycle} - (\text{number of phases} \times \text{intergreen})$$

the time left for traffic to move

$$\text{green}_i = \text{total green} \times \text{split share}_i$$

each phase's green

SIMPLE CALCULATION · Building a 90 s cycle

Two phases, 5 s intergreen each, split 60% to A and 40% to B.

1. Lost time = 2 × 5 = **10 s**.
2. Total green = 90 − 10 = **80 s**.
3. Phase A green = 80 × 0.60 = **48 s**; phase B green = 80 × 0.40 = **32 s**.
4. Share of the cycle lost = 10 ÷ 90 = **11.1%**. With a 60 s cycle it would be 10 ÷ 60 = **16.7%**.

Longer cycles waste a smaller share of time, but make drivers wait longer at red. Choosing the cycle is a balance.

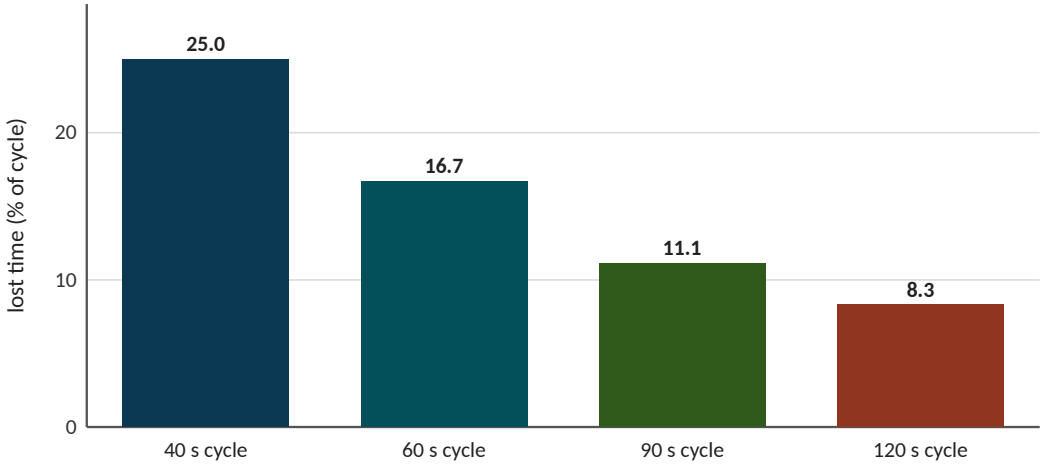


Figure 3. Share of the cycle lost to intergreens (2 phases × 5 s). What to notice: the same 10 s of intergreen is a quarter of a 40 s cycle but under a tenth of a 120 s cycle. Computed as 10 ÷ cycle.

TRAP

The cycle includes the intergreens. A 90 s cycle with two 5 s intergreens has only 80 s of green to share, not 90.

PICTURE THIS · STOP 3 OF THE ROUTE

On a tower above the bridge, the clock face is a giant pizza. The hand sweeps round once per cycle. Two slices, one big and one small, are painted A and B, and a thick burnt crust between them crunches loudly every time the hand passes.

What it encodes: Whole pizza = cycle; slices = phases; slice sizes = split; burnt crust = intergreen (lost time); the hand's start time vs the next tower = offset.

TRY IT NOW (about 90 seconds)

A 120 s cycle has three phases with 5 s intergreen each and a split of 50% / 30% / 20%. Find each green and the share of the cycle lost. *Answer in the key at the back.*

RECAP SO FAR

1. **What SCATS is:** adaptive (every cycle), coordinated (shared cycle, offsets) and plan-based (approved plans); it grades each green and nudges the next cycle.
2. **Two levels:** tactical control every second in the local controller (gap-out, skip); strategic control every cycle in the regional computer (cycle towards DS 0.9, splits, offsets).
3. **Timing words: phase = movements together; cycle = all phases plus intergreens; split = share of green; intergreen = amber + all-red (lost); offset = start time vs neighbour.**

CONCEPT 4 OF 6 · SCATS FOUNDATION

Measuring demand: the degree of saturation

HOOK

How do you know how busy a green was, without counting cars? Measure the empty road. Whatever was not empty was used.



IN PLAIN WORDS

Think of a supermarket checkout. Even when the queue is endless there is a short pause between customers. Any pause longer than that normal pause is the cashier waiting for nobody: wasted time. *Where the picture breaks:* a cashier's normal pause varies by person; SCATS uses one standard value h per lane, which must be calibrated.

During each green SCATS watches the stop-line loop and adds up the **gap time T**: every moment when nothing is over the loop. It also counts the **vehicles n**.

Even a perfectly packed stream of cars leaves a small gap between each pair, the **standard space time h**. So n vehicles 'should' leave n × h of gap. Any gap beyond that, T − n·h, is **wasted green**. The rest is **used green g′**, and the **degree of saturation** is the share of the green that was used.

SCATS also turns DS into **VK**, a car-equivalent volume, using **MF**, the maximum flow the lane can carry (learned automatically). Because DS is measured in time, a truck that covers the loop longer counts for more, just as it uses more green.

$$g' = g - (T - n{\cdot}h)$$

used green = green minus wasted gap time

$$DS = g' \div g$$

1 means the green was fully used

$$VK = DS \times g \times MF \div 3600$$

car-equivalent volume; MF in vehicles per hour of green

SIMPLE CALCULATION · DS and VK from one green

Green g = 40 s. The loop was empty for T = 20 s. n = 10 vehicles crossed. h = 1.6 s, MF = 1800 veh/h.

1. Expected gaps: n × h = 10 × 1.6 = **16 s**.
2. Wasted green: T − n·h = 20 − 16 = **4 s**.
3. Used green: g′ = 40 − 4 = **36 s**.
4. DS = 36 ÷ 40 = **0.90**.
5. VK = 0.90 × 40 × 1800 ÷ 3600 = **18** car-equivalents.

90% of the green was used: right at the usual target. If h were set wrongly, DS would be wrong too, which is why h is calibrated.

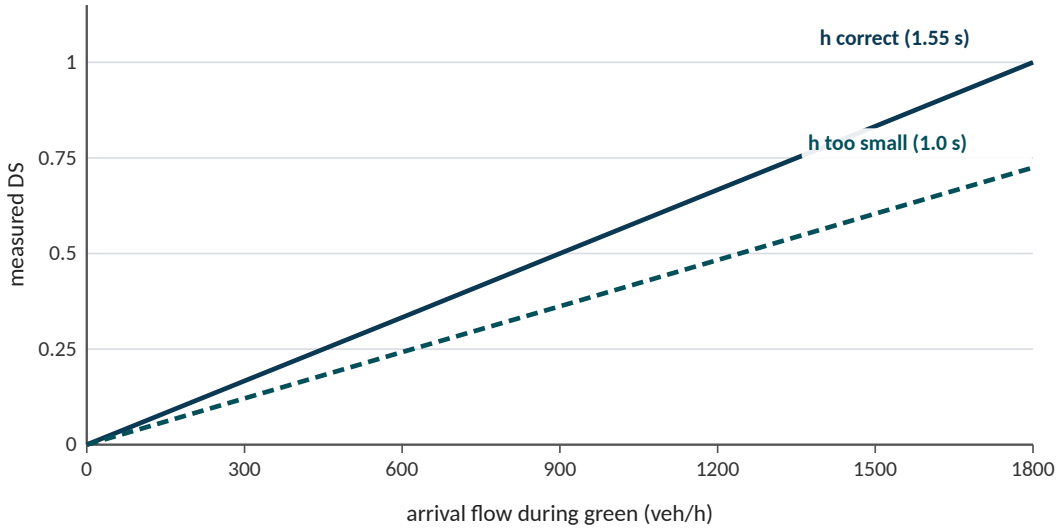


Figure 4. Measured DS against true arrival flow, for a correct and a wrong h. What to notice: with the true h (1.55 s here: one car every 2 s, each covering the loop 0.45 s) DS rises in a straight line to 1 at 1800 veh/h. With h set too small (1.0 s) DS is under-estimated by about a quarter, so SCATS would give too little green. Computed for g = 40 s.

TRAP

DS is not 'number of cars ÷ capacity' counted in vehicles. It is measured in time from the loop's gaps, and it depends on h. A wrong h gives a wrong DS for every cycle.

PICTURE THIS · STOP 4 OF THE ROUTE

By the bridge's far pylon an ice-cream seller serves a queue. Between customers she holds up a ruler 1.6 long; any gap longer than the ruler, she drops a cone in a bin labelled WASTED. A scoreboard shows 'used ÷ open time'.

What it encodes: Ruler = h (1.6 s); cones in the bin = wasted green T − n·h; the scoreboard = DS = g′ ÷ g.

TRY IT NOW (about 90 seconds)

Green g = 30 s, gap time T = 15 s, n = 8 vehicles, h = 1.6 s, MF = 1730 veh/h. Find DS and VK. *Answer in the key at the back.*

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4. **Degree of saturation: wasted = T − n·h, used g′ = g − wasted, DS = g′/g, VK = DS·g·MF/3600; h must be calibrated.**

CONCEPT 5 OF 6 · SCATS FOUNDATION

Fixed time, actuated and SCATS

HOOK

Put three controllers on the same junction with the same cars. The plain timer loses badly. But the race between the other two is much closer than you might expect.



IN PLAIN WORDS

Fixed time is a lift that stops at every floor whether anyone is waiting or not. Actuated is a lift that skips empty floors. SCATS also skips empty floors, and every few minutes rethinks its whole timetable from how busy it was. *Where the picture breaks:* one lift cannot show coordination; SCATS's biggest advantage comes from many intersections working together (Advanced level).

Fixed time runs one plan (here a 90 s cycle) whatever the traffic. **Actuated** control uses the loops to end greens early by gap-out but keeps fixed maximum greens. **SCATS** adds the strategic layer: after every cycle it computes DS, moves the cycle length towards DS 0.9 and votes on the best of four split plans.

In Lab F3 (one intersection in SUMO, one hour with a peak in the middle, identical arrivals) the mean delay was **20.44 s** for fixed time, **14.96 s** for actuated and **14.27 s** for SCATS. Stops per vehicle were 0.64, 0.62 and 0.565.

At one isolated intersection SCATS and good actuation finish close together: both end greens when traffic clears. SCATS earns its keep through coordinating many intersections.

$$\text{improvement} = (\text{old} - \text{new}) \div \text{old} \times 100\%$$

how we compare controllers fairly

SIMPLE CALCULATION · How big are the differences?

Lab F3 mean delays: fixed 20.44 s, actuated 14.96 s, SCATS 14.27 s.

1. SCATS vs fixed: (20.44 − 14.27) ÷ 20.44 = 6.17 ÷ 20.44 = **30.2%** less delay.
2. SCATS vs actuated: (14.96 − 14.27) ÷ 14.96 = 0.69 ÷ 14.96 = **4.6%** less delay.
3. Actuated vs fixed: (20.44 − 14.96) ÷ 20.44 = **26.8%** less delay.

Most of the gain over fixed time comes from reacting to gaps; SCATS's strategic layer adds a little more at a single intersection.

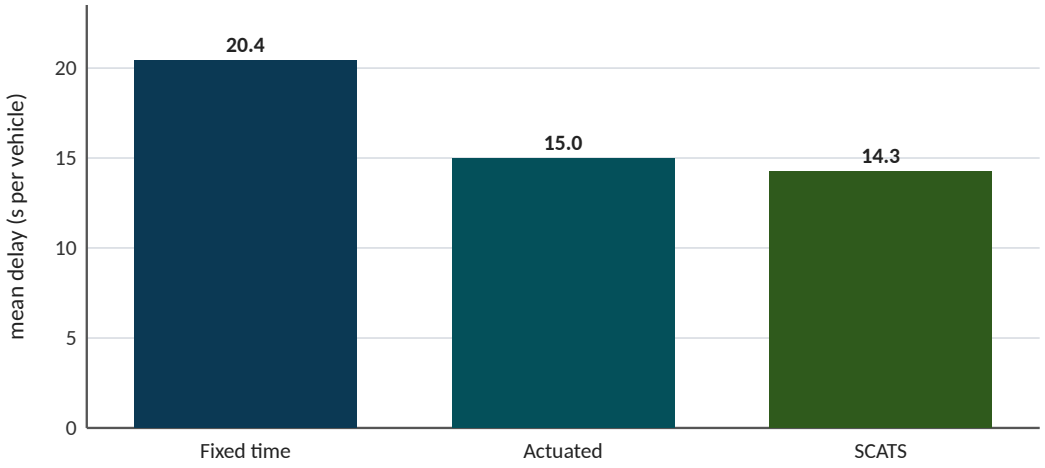


Figure 5. Mean delay per vehicle in Lab F3 (SUMO, one intersection, one hour). What to notice: fixed time is clearly worst; actuated and SCATS are close, with SCATS slightly ahead. Results from the repository's SUMO run (seed 1).

TRAP

Do not conclude 'SCATS is only 5% better than actuation'. This test is one isolated intersection. Coordination across a corridor, which actuation cannot do, is where the bigger gains appear.

PICTURE THIS · STOP 5 OF THE ROUTE

Below the bridge three ferries race to the quay. A rusty clockwork ferry chugs in at twenty and a half. A jumpy speedboat arrives at fifteen, and just ahead of it, at fourteen and a quarter, glides a ferry whose captain studies a chart after every lap.

What it encodes: Finish times are Lab F3 delays: fixed 20.4 s, actuated 15.0 s, SCATS 14.3 s; the chart-reading captain = SCATS's strategic layer.

TRY IT NOW (about 90 seconds)

Stops per vehicle in Lab F3 were 0.64 (fixed) and 0.565 (SCATS). By what percentage did SCATS cut stops? *Answer in the key at the back.*

RECAP SO FAR

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3. **Timing words:** phase = movements together; cycle = all phases plus intergreens; split = share of green; intergreen = amber + all-red (lost); offset = start time vs neighbour.
4. **Degree of saturation:** wasted = T − n·h, used g′ = g − wasted, DS = g′/g, VK = DS·g·MF/3600; h must be calibrated.
5. **Three controllers: fixed ignores traffic, actuated gaps out, SCATS gaps out and re-plans each cycle. Lab F3 delay: 20.4 / 15.0 / 14.3 s.**

CONCEPT 6 OF 6 · SCATS FOUNDATION

Where fixed-time plans come from

HOOK

A fixed plan is a suit tailored for one day's traffic. Webster's formula is the tailor. SCATS, in effect, re-measures the customer every cycle.



IN PLAIN WORDS

Webster is a tailor who measures you once and sews a perfect suit. It fits beautifully on that day. If you gain weight or wear a coat underneath, the suit no longer fits, and nobody re-measures you. *Where the picture breaks:* a suit cannot change shape; SCATS re-solves the timing problem every cycle from measured DS, so its 'suit' keeps fitting.

Fixed-time plans are usually designed with **Webster's formula** (1958). It balances lost time against queueing for one assumed demand. Each phase's **flow ratio** y is its flow q divided by its saturation flow s; Y adds up the critical ones; L is the total lost time per cycle.

The formula gives the cycle with the least delay, and shares the effective green in proportion to the flow ratios. As Y approaches 1 the cycle grows without limit: the intersection is near capacity.

SCATS needs no design day, but it does need calibration. Lab F2 overloaded the main road and measured **h = 1.62 s** and a maximum flow **MF ≈ 1730 veh/h**, the values the controller uses.

$$C_0 = (1.5\,L + 5) \div (1 - Y)$$

L = total lost time (s), Y = sum of critical flow ratios q/s

$$g_i = (C_0 - L) \times y_i \div Y$$

effective green shared by flow ratio

SIMPLE CALCULATION · Webster for a two-phase intersection

Lost time L = 10 s. Main road y_A = 0.40, side road y_B = 0.20.

1. Y = 0.40 + 0.20 = **0.60**.
2. C_0 = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = **50 s**.
3. Green to share = 50 − 10 = **40 s**.
4. Main road: 40 × 0.40 ÷ 0.60 = **26.7 s**. Side road: 40 × 0.20 ÷ 0.60 = **13.3 s**.

A good fixed plan for exactly these flows: 50 s cycle, 26.7 s and 13.3 s of effective green.

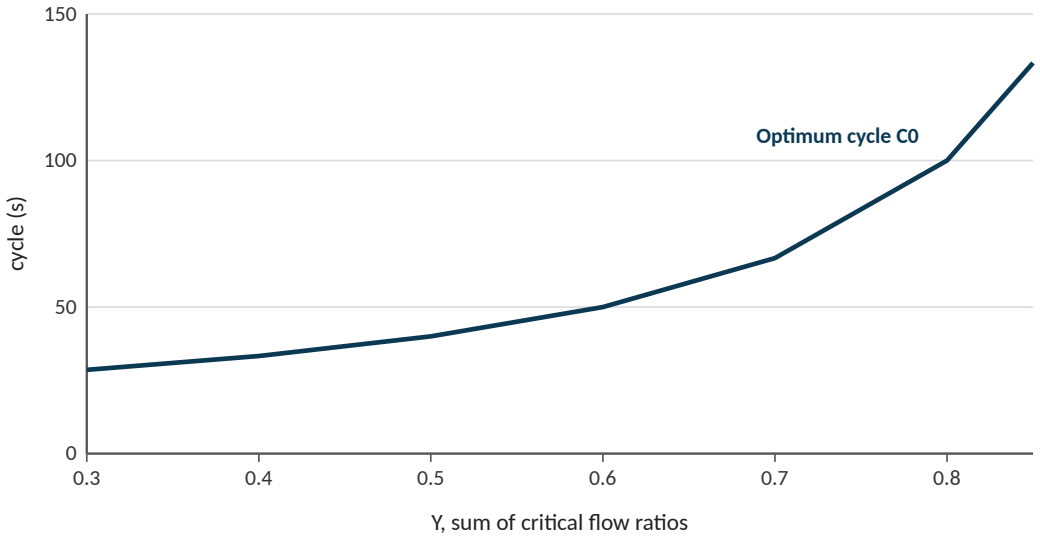


Figure 6. Webster's optimum cycle as demand grows (L = 10 s). What to notice: the cycle grows slowly at first, then shoots up as Y nears 1. At Y = 0.85 it is already 133 s. Computed from $C_0 = 20 \div (1 - Y)$.

TRAP

If Y ≥ 1 no cycle length works: demand exceeds capacity. A huge cycle from Webster is a warning that the intersection is near capacity, not a plan to use.

PICTURE THIS · STOP 6 OF THE ROUTE

On the opera house steps a tailor stitches a suit onto a dancer. His tape measure is printed '1.5 L + 5', and the suit fits perfectly until the dancer eats a giant pie and the buttons fly off. Next door an assistant re-measures a second dancer after every pirouette, holding a ruler marked 1.62.

What it encodes: Tailor = Webster's formula $C_0 = (1.5L + 5)/(1 - Y)$ for one design day; popping buttons = demand changing; the re-measuring assistant = SCATS re-solving every cycle; 1.62 = calibrated h.

TRY IT NOW (about 90 seconds)

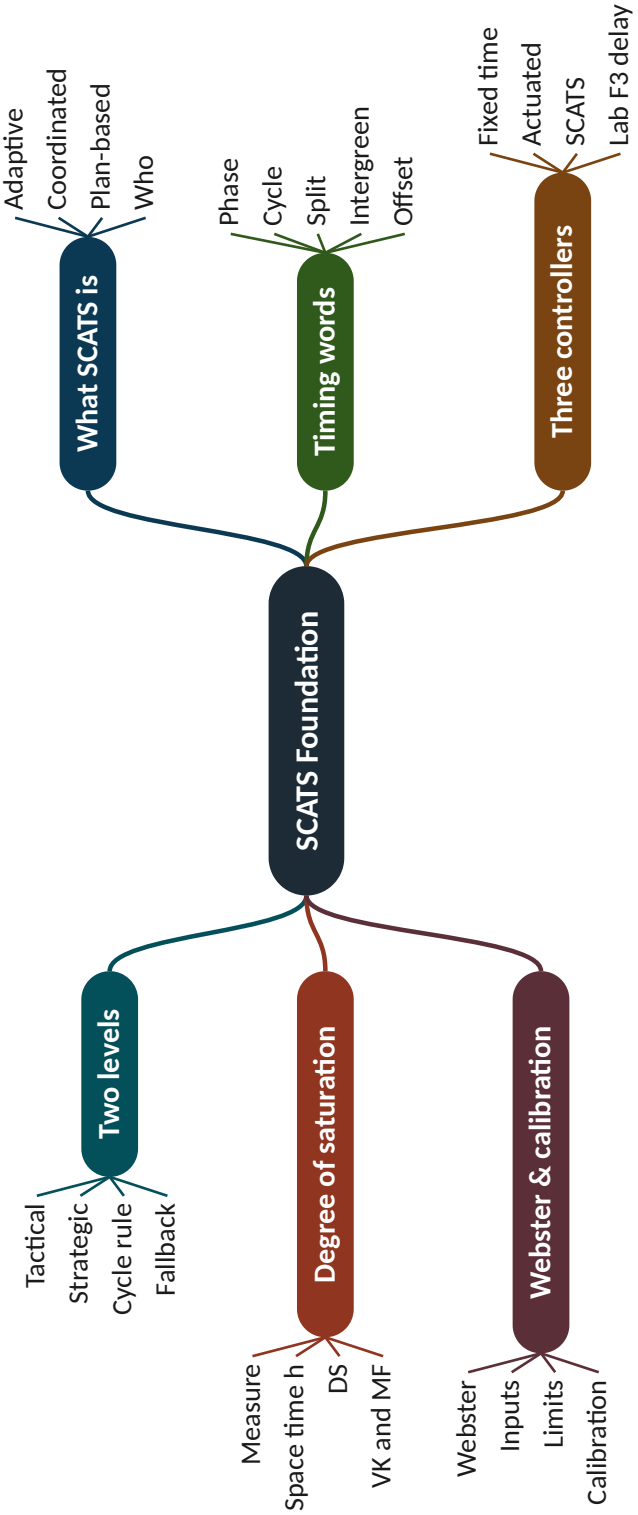
Use Webster with L = 12 s, $y_A = 0.45$, $y_B = 0.25$. Find C_0 and both greens. *Answer in the key at the back.*

RECAP SO FAR

1. **What SCATS is:** adaptive (every cycle), coordinated (shared cycle, offsets) and plan-based (approved plans); it grades each green and nudges the next cycle.
2. **Two levels:** tactical control every second in the local controller (gap-out, skip); strategic control every cycle in the regional computer (cycle towards DS 0.9, splits, offsets).
3. **Timing words:** phase = movements together; cycle = all phases plus intergreens; split = share of green; intergreen = amber + all-red (lost); offset = start time vs neighbour.
4. **Degree of saturation:** wasted = T − n·h, used g′ = g − wasted, DS = g′/g, VK = DS·g·MF/3600; h must be calibrated.
5. **Three controllers:** fixed ignores traffic, actuated gaps out, SCATS gaps out and re-plans each cycle. Lab F3 delay: 20.4 / 15.0 / 14.3 s.
6. **Webster and calibration: $C_0 = (1.5L + 5)/(1 - Y)$ designs one fixed plan; SCATS re-solves every cycle but needs calibrated h (1.62 s) and MF (≈ 1730 veh/h).**

The master mind map

Every concept and its main branches on one page. Turn the page sideways. Try redrawing it from memory on blank paper, then compare.



Connections and contrasts

Three ways to run a traffic light

	Fixed time	Actuated	SCATS
Detectors	none needed	stop-line loops	stop-line loops in every lane
Ends a green early?	no	yes, by gap-out	yes, by gap-out
Changes the cycle?	no	no (fixed maximum)	every cycle, towards DS 0.9
Coordinates neighbours?	if designed so	no	yes: subsystems, offsets
Lab F3 delay	20.44 s	14.96 s	14.27 s

Things beginners mix up

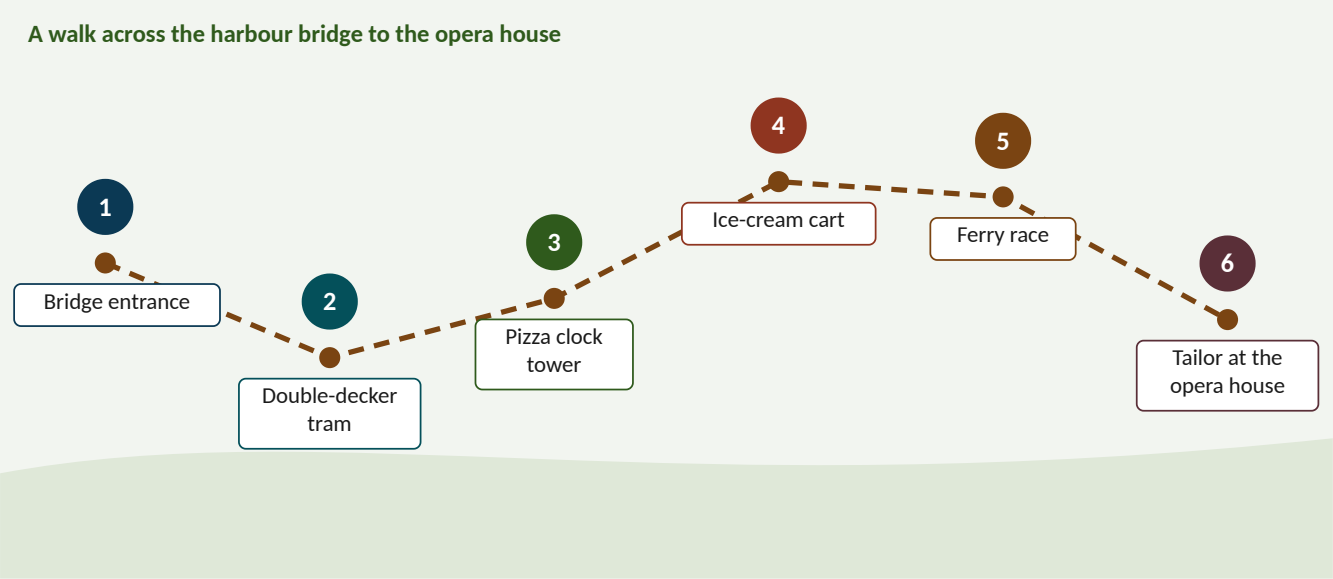
This…	…is not this	The one fact that tells them apart
Tactical	**Strategic**	Tactical decides every second in the local controller; strategic decides once per cycle in the regional computer.
Cycle	**Total green**	The cycle includes the intergreens; total green is what is left after them.
Gap time T	**Wasted green**	Wasted green is T minus the normal gaps n·h.
DS	**Vehicle count ÷ capacity**	DS is measured in time from loop gaps, so trucks count for more automatically.
Adaptive	**Unlimited**	SCATS changes timings in small steps and from approved plans.

Symbols and terms

Symbol / term	Meaning
g	green time available to the movement (s)
T	total time the loop was empty during that green (s)
n	vehicles detected during the green
h	standard space time: gap per vehicle at saturation (s)
g′	used green = g − (T − n·h)
DS	degree of saturation = g′ ÷ g
MF, VK	maximum flow (veh/h of green); car-equivalent volume DS·g·MF/3600
L, Y, C_0	Webster's lost time, sum of flow ratios, optimum cycle

The route: A walk across the harbour bridge to the opera house

Picture yourself walking across Sydney's Harbour Bridge towards the Opera House. Six things happen on the way, one for each concept. Make every scene loud, silly and physical.



The memory route. Walk it in order; each stop holds one concept.

Stop 1: Bridge entrance (What SCATS is). At the start of the bridge a conductor in a huge coat waves a baton at the traffic lights. Instead of one score she flips between four thick music books, and after every bar she scribbles a red mark on the page she just played. *Conductor = coordinated; red marks after every bar = adaptive grading of each cycle; four music books = the library of split plans (plan-based).*

Stop 2: Double-decker tram (Two levels of control). Halfway across the bridge a double-decker tram rattles by. Downstairs a driver slaps a big button every single second, shouting 'empty, next!'. Upstairs a captain with a clipboard looks through a telescope and, once a lap, turns a dial by exactly six clicks. *Downstairs driver = tactical control every second (gap-out, skip); upstairs captain = strategic control once per cycle; six clicks = the ±6 s cycle step.*

Stop 3: Pizza clock tower (The timing vocabulary). On a tower above the bridge, the clock face is a giant pizza. The hand sweeps round once per cycle. Two slices, one big and one small, are painted A and B, and a thick burnt crust between them crunches loudly every time the hand passes. *Whole pizza = cycle; slices = phases; slice sizes = split; burnt crust = intergreen (lost time); the hand's start time vs the next tower = offset.*

Stop 4: Ice-cream cart (Measuring demand: the degree of saturation). By the bridge's far pylon an ice-cream seller serves a queue. Between customers she holds up a ruler 1.6 long; any gap longer than the ruler, she drops a cone in a bin labelled WASTED. A scoreboard shows 'used ÷ open time'. *Ruler = h (1.6 s); cones in the bin = wasted green $T - n \cdot h$; the scoreboard = $DS = g' \div g$.*

Stop 5: Ferry race (Fixed time, actuated and SCATS). Below the bridge three ferries race to the quay. A rusty clockwork ferry chugs in at twenty and a half. A jumpy speedboat arrives at fifteen, and just ahead of it, at fourteen and a quarter, glides a ferry whose captain studies a chart after every lap. *Finish times are Lab F3 delays: fixed 20.4 s, actuated 15.0 s, SCATS 14.3 s; the chart-reading captain = SCATS's strategic layer.*

Stop 6: Tailor at the opera house (Where fixed-time plans come from). On the opera house steps a tailor stitches a suit onto a dancer. His tape measure is printed '1.5 L + 5', and the suit fits perfectly until the dancer eats a giant pie and the buttons fly off. Next door an assistant re-measures a second dancer after every pirouette, holding a ruler marked 1.62. *Tailor = Webster's formula $C_0 = (1.5L + 5)/(1 - Y)$ for one design day; popping buttons = demand changing; the re-measuring assistant = SCATS re-solving every cycle; 1.62 = calibrated h.*

> **WALK-THROUGH DRILL**
>
> Close the guide. Walk the route in your head, saying each stop out loud and the one number or formula it holds. Then check against the cheat sheet. Repeat until you can do it in under two minutes.

Self-quiz

18 questions: warm-up (recall), core (use the ideas), challenge (combine them). Answer on paper first, then check the key.

Warm-up

Q1. What does the 'A' in SCATS stand for?

A. Automatic
B. Adaptive
C. Actuated
D. Arterial

Q2. Where are SCATS's detectors normally placed?

A. Far upstream of the junction
B. At the stop line in every lane
C. On the signal heads
D. Only on the main road

Q3. Which decision is tactical?

A. Choosing the cycle length
B. Voting on a split plan
C. Ending a green early because the road emptied
D. Choosing offsets

Q4. What is the intergreen?

A. The green between two reds
B. Amber plus all-red between phases
C. The offset between intersections
D. The minimum green

Q5. A DS of 1.0 means…

A. the road is empty
B. the green was fully used
C. the signal failed
D. half the green was wasted

Core

Q6. A 90 s cycle has two phases with 5 s intergreen each, split 70/30. What is phase A's green?

A. 63 s
B. 56 s
C. 59.5 s
D. 45 s

Q7. g = 40 s, T = 12 s, n = 5, h = 1.6 s. What is DS?

A. 0.50
B. 0.70
C. 0.90
D. 0.30

Q8. If h is set too small, DS will be…

A. over-estimated
B. under-estimated
C. unchanged
D. always exactly 1

Q9. DS = 0.8, g = 30 s, MF = 1800 veh/h. What is VK?

A. 12
B. 24
C. 6.7
D. 48

Q10. The cycle is 100 s and the busiest phase had DS 0.72. Using this lab's rule, what is the next cycle?

A. 80 s
B. 94 s
C. 106 s
D. 100 s

Q11. Why does a longer cycle waste a smaller share of time?

A. Intergreens get shorter
B. The same intergreens are spread over more time
C. Drivers drive faster
D. Gap-out stops working

Q12. Which fallback runs time-of-day plans with clocks kept in step?

A. Masterlink
B. Flexilink
C. Isolated
D. Gap-out

Q13. Webster with L = 10 s and Y = 0.5. What is C_0?

A. 30 s
B. 40 s
C. 50 s
D. 60 s

Q14. In Lab F3, which controller had the lowest mean delay?

A. Fixed time
B. Actuated
C. SCATS
D. All equal

Challenge

Q15. Two lanes on the same phase: lane 1 DS 0.95, lane 2 DS 0.60. Which DS should represent the phase, and why?

A. The average, 0.775
B. The highest, 0.95, because the busiest lane decides if the green is enough
C. The lowest, 0.60
D. Neither; DS is per intersection

Q16. At one isolated intersection SCATS beat actuation by only 4.6%. What is the best conclusion?

A. SCATS is not worth it
B. Both react to gaps; SCATS's bigger gains come from coordination
C. The simulation is broken
D. Actuation is always better

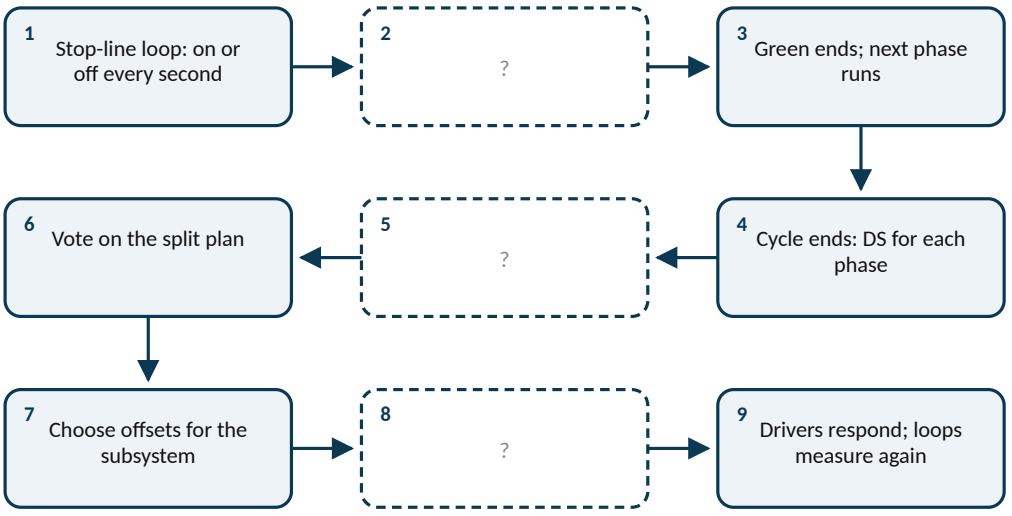
Q17. Webster gives C_0 = 400 s for a busy junction. What should an engineer conclude?

A. Use 400 s
B. Y is near 1: capacity, not timing, is the problem
C. Webster only works for small junctions
D. Halve the lost time

Q18. A truck and a car pass the loop during a saturated green. How does DS treat the truck?

A. Ignores it
B. Counts it as one car
C. It leaves less empty time, so it raises DS as it uses more green
D. It lowers DS

Draw it. Fill the blank boxes of this flow from memory (it is Figure 1).



Where to go if you missed a question

Missed	Do this
Q1	Re-read concept 1: What SCATS is
Q3, Q10, Q12	Re-read concept 2: Two levels of control
Q4, Q6, Q11	Re-read concept 3: The timing vocabulary
Q2, Q5, Q7, Q8, Q9, Q15, Q18	Re-read concept 4: Measuring demand: the degree of saturation
Q14, Q16	Re-read concept 5: Fixed time, actuated and SCATS
Q13, Q17	Re-read concept 6: Where fixed-time plans come from

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 min)

On paper, take a 40 s green. 8 vehicles crossed and the loop was empty for 22 s in total. With h = 1.6 s, compute DS. Then imagine 11 vehicles cross and the empty time drops to 19 s; recompute DS and say whether SCATS would want a longer or shorter cycle.

Self-check: First case: n·h = 12.8 s, wasted 9.2 s, g′ = 30.8 s, DS = 0.77. Second: n·h = 17.6 s, wasted 1.4 s, g′ = 38.6 s, DS = 0.965, above 0.9, so SCATS would lengthen the cycle.

TEACH IT TO A 14-YEAR-OLD (10 min)

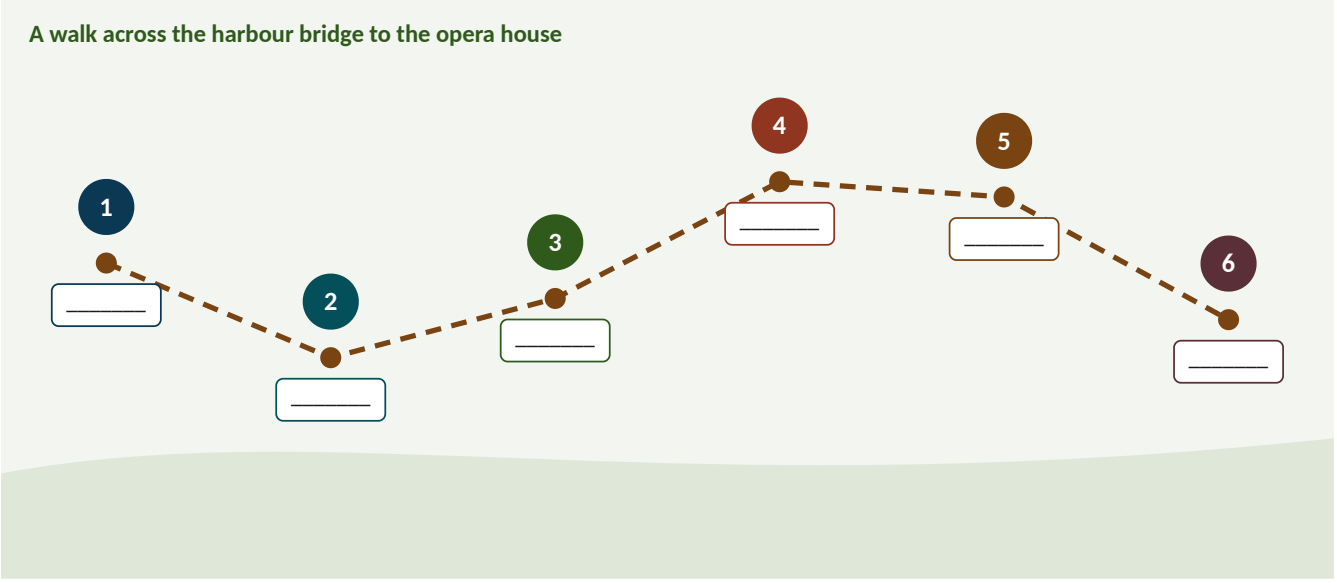
Explain to a 14-year-old how a traffic light can tell how busy the road was by watching the empty moments on one patch of road, and why it changes its timing only a little each time.

Words you may not use: saturation, tactical, strategic, intergreen, phase

Fill the blank route

Write the concept held at each stop, and its key number.

A walk across the harbour bridge to the opera house



One-page cheat sheet

1. What SCATS is

Sydney Coordinated Adaptive Traffic System, Transport for NSW, from the 1970s. Adaptive + coordinated + plan-based. Grades each cycle, nudges the next. 90 s cycle → 40 decisions/h.

Route: Bridge entrance

2. Two levels of control

Tactical: local, every second, gap-out (3 s), skip, priority. Strategic: regional, every cycle, cycle/split/offset per subsystem. Desired C = C × DS/0.9, step +6 (+9 if DS ≥ 1)/ −6, 40-140 s. Fallback: Masterlink → Flexilink → Isolated.

Route: Double-decker tram

3. The timing vocabulary

Green = C − phases × intergreen; g_i = green × $share_i$. 90 s, 2 × 5 s, 60/40 → 48 s / 32 s. Lost share 10/C: 25% at 40 s, 11.1% at 90 s.

Route: Pizza clock tower

4. Measuring demand: the degree of saturation

$g' = g - (T - n \cdot h)$; DS = g'/g; VK = DS·g·MF/3600. Example g 40, T 20, n 10, h 1.6 → DS 0.90, VK 18. h too small → DS too low.

Route: Ice-cream cart

5. Fixed time, actuated and SCATS

Fixed 90 s plan; actuated = gap-out + fixed max; SCATS = gap-out + cycle to DS 0.9 + split vote. Lab F3 delay 20.44 / 14.96 / 14.27 s; stops 0.64 / 0.62 / 0.565. SCATS −30% vs fixed, −4.6% vs actuated.

Route: Ferry race

6. Where fixed-time plans come from

$C_0 = (1.5L+5)/(1-Y)$; $g_i = (C_0 - L) \cdot y_i / Y$. L 10, Y 0.6 → 50 s (26.7 / 13.3 s). Y ≥ 1 → over capacity. Lab F2: h = 1.62 s, MF ≈ 1730 veh/h.

Route: Tailor at the opera house

Numbers to remember

- Target DS ≈ 0.9
- Cycle 40-140 s, step +6 (+9 if DS ≥ 1) / −6 s
- Gap-out setting 3 s; intergreen 5 s
- h = 1.62 s, MF ≈ 1730 veh/h (Lab F2)
- g 40, T 20, n 10, h 1.6 → DS 0.90, VK 18
- Lab F3 delay: 20.44 / 14.96 / 14.27 s
- Webster: L 10, Y 0.6 → C_0 = 50 s

Answer key

Try it now

Try it 1 (What SCATS is): 3600 ÷ 120 = **30** cycles, so **30** strategic decisions (one per cycle).

Try it 2 (Two levels of control): Desired = 60 × 1.05 ÷ 0.9 = **70 s**, 10 s longer. DS ≥ 1, so the step limit is +9 s: new cycle = **69 s**.

Try it 3 (The timing vocabulary): Lost = 3 × 5 = 15 s; green = 105 s. Greens: **52.5 s**, **31.5 s**, **21 s**. Lost share = 15 ÷ 120 = **12.5%**.

Try it 4 (Measuring demand: the degree of saturation): n·h = 12.8 s; wasted = 15 − 12.8 = 2.2 s; g′ = 27.8 s; DS = 27.8 ÷ 30 = **0.927**. VK = 0.927 × 30 × 1730 ÷ 3600 = **13.4**.

Try it 5 (Fixed time, actuated and SCATS): (0.64 − 0.565) ÷ 0.64 = 0.075 ÷ 0.64 = **11.7%** fewer stops.

Try it 6 (Where fixed-time plans come from): Y = 0.70. C_0 = (18 + 5) ÷ 0.30 = **76.7 s**. Green to share = 64.7 s. Main 64.7 × 0.45/0.70 = **41.6 s**; side 64.7 × 0.25/0.70 = **23.1 s**.

Self-quiz

Q1: B. Sydney Coordinated Adaptive Traffic System. Actuated control is a different, simpler idea. (Concept 1)

Q2: B. SCATS uses stop-line loops in every lane; far-upstream loops are SCOOT's approach. (Concept 4)

Q3: C. Gap-out happens every second in the local controller. The others are strategic, once per cycle. (Concept 2)

Q4: B. Amber plus all-red: time lost to traffic between phases. (Concept 3)

Q5: B. DS = used green ÷ green; 1 means all of it was used. (Concept 4)

Q6: B. Green = 90 − 10 = 80 s; 80 × 0.70 = 56 s. 63 s forgets to remove the intergreens. (Concept 3)

Q7: C. n·h = 5 × 1.6 = 8 s; wasted = 12 − 8 = 4 s; g′ = 40 − 4 = 36 s; DS = 36 ÷ 40 = 0.90. 0.70 forgets the normal gaps. (Concept 4)

Q8: B. A small h treats normal gaps as wasted green, so used green and DS come out too low. (Concept 4)

Q9: A. VK = 0.8 × 30 × 1800 ÷ 3600 = 12. (Concept 4)

Q10: B. Desired = 100 × 0.72 ÷ 0.9 = 80 s, but the step is limited to −6 s, giving 94 s. (Concept 2)

Q11: B. Lost time per cycle stays roughly fixed, so its share falls as the cycle grows. (Concept 3)

Q12: B. Flexilink. Masterlink is full adaptive control; Isolated is local actuation. (Concept 2)

Q13: B. C_0 = 20 ÷ 0.5 = 40 s. 50 s is the Y = 0.6 example. (Concept 6)

Q14: C. SCATS 14.27 s, actuated 14.96 s, fixed 20.44 s. (Concept 5)

Q15: B. Like the lab's controller, take the highest DS among the phase's loops: if the busiest lane runs short, the green is not enough. (Concept 4)

Q16: B. Most of the single-junction gain is gap-out, which both have. Coordination is tested at the Advanced level. (Concept 5)

Q17: B. As Y → 1 the cycle explodes; controllers cap it and the real issue is capacity. (Concept 6)

Q18: C. DS is measured in time: a long vehicle covers the loop longer, leaving less gap time. (Concept 4)

Draw it: the completed flow is Figure 1.

Your revision plan

When	What	Done
Today	Read one concept a sitting. Do every Try-it-now on paper.	☐
Day 1	Walk the route from memory. Redraw the master mind map.	☐
Day 3	Warm-up and core quiz without notes. Re-read what you missed.	☐
Day 7	Whole quiz. Do the 'build the smallest example' task.	☐
Day 14	Teach-it task out loud. Cheat sheet from memory, then compare.	☐
Before an exam or project	Cheat sheet plus route walk only.	☐

Going further

- Sims, A.G. and Dobinson, K.W. (1980). The Sydney Coordinated Adaptive Traffic (SCAT) System philosophy and benefits. *IEEE Transactions on Vehicular Technology*, 29(2), 130-137.
- Lowrie, P.R. (1982). The Sydney Coordinated Adaptive Traffic System: principles, methodology, algorithms. *IEE Conference Publication 207*, 67-70.
- Webster, F.V. (1958). *Traffic signal settings*. Road Research Technical Paper No. 39.

- This lab's website: Foundation page and SUMO labs F1-F3, smrahmanbd.github.io/scats-learning-lab.

HONESTY NOTE

SCATS is owned by Transport for NSW; this guide is an independent, educational explanation of concepts published in the literature. Worked examples use small, invented numbers. Parameter values (target DS 0.9, ±6 s steps, 3 s gap, h and MF) are this lab's illustrative teaching values, not official SCATS settings. Lab F1-F3 results come from this repository's SUMO runs (seed 1).